Suraj Kumar

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EDUCATION

University of Delaware — Graduation: **May 2029**
Ph.D. in **Computer Science** — GPA: **4.00/4.00**

University of South Florida — Graduation: **May 2024**
Bachelor of Science in **Computer Science** — GPA: **3.79/4.00**

SKILLS

Software and Tools: Python, SQL, JavaScript, PyTorch, OpenCV, SAS, Git, TensorBoard, Scikit-learn, Seaborn, FastAPI, LaTeX
Core Competencies: Deep Learning (CNNs, Attention-based Models), Computer Vision, Data Analytics, Statistical Modeling

WORK EXPERIENCE

Research Assistant (EHR, Transformers) **–** *Healthy LAIfe Lab, Newark, DE* — **Feb 2026 – Present**
- Developing age-conditioned attention mechanisms for pediatric EHR data, comparing architectural conditioning strategies (additive injection vs. FiLM-style) for developmental risk detection
- Benchmarking longitudinal trajectory encoders (**HyMaTE, Mamba-based**) on PIC dataset for clinical prediction tasks

Research Intern (ML, Computer Vision) **–** *Nemours Children's Hospital, Wilmington, DE* — **Jul 2025 – Dec 2025**
- Developed **ML** based OCT analysis pipeline for Sickle Cell Retinopathy, detecting **>50%** retinal lesions than standard FA exams
- Trained and evaluated deep learning segmentation models (**Encoder-Decoder**, **U-Net**; achieved **98.6%** agreement with expert annotations) and object detection models (**RCNN, YOLOv3**) on topographic retinal maps to accurately detect key biomarkers

Research Assistant (NLP, GenAI) **–** *AHMR Lab, Tampa, FL* — **Aug 2024 – Jul 2025**
- Built multi-agent RISK simulation using persona-based heuristics via **Ollama**, improving AI decision accuracy
- Designed and benchmarked **RAG** prompting techniques (**CoT**, **Few-shot**, **LOCI**), identifying methods that improved rule-based inference accuracy by **15%**

Teaching Assistant – *CS Department, USF, Tampa, FL* — **Aug 2023 – May 2024**
- Supported **50+** students per semester with assignments and projects, contributing to a marked improvement in overall class performance
- Conducted weekly office hours and led peer programming sessions to support student understanding

Automation Intern – *Nucor Steel Texas, Jewett, TX* — **May 2023 – Aug 2023**
- Developed a Power App to effectively monitor electricity costs, using **Power FX** and **SQL** database, including functionalities like confirmation popup, sound alert and real time electricity price which helped decrease the data reload time by **12%**
- Engineered **ML**-based scrap recipe optimization system, using **Python**, **SQL** and **JavaScript**, predicting chemistry and lowering production cost by **5–10%**

PUBLICATIONS

"**Steps Towards ML-Assisted Clinical Staging of Retinal Disease**," **Kumar S.**, et al. link — **Oct 2025**
Poster Presented at Delaware Data Science Symposium 2025

"**Curiosity Exploration Styles in Word Association Task**" – *Steinle S., **Kumar S.**, Licato J., et al.* link — **Jun 2025**
Peer-reviewed paper published in CogSci 2025

"**Improving Multi-hop Logical Reasoning in Small LMs with LoRA Training**" – *Bilgin O., **Kumar S.**, Licato J., et al.* link — **May 2025**
Peer-reviewed paper published in FLAIRS38

LEADERSHIP

Co-Director – *Hackabull* — **May 2022 – May 2023**
- Organized and steered Tampa's largest hackathon, overseeing a team of **12**, implemented effective project management techniques which resulted in a successful event with **500+** attendees, a **55%** increase from the previous year

VP of Communications – *SHPE USF* — **Mar 2022 – May 2023**
- Designed and distributed weekly newsletters curated to meet the needs of over **1100** STEM students, providing information about upcoming events and employment opportunities resulting in greater student engagement by **22%**